

Programming Language Foundations Appunti

Alex "Morbidelli WhatsApp" Basta

Contents

Chapter 1

Inductive Definitions and Proofs

Page

- 1.1 Properties of programs
 - Inductive Sets — • Inductive Proofs —

Chapter 1

Inductive Definitions and Proofs

Definition 1.0.1: Multi-step evaluation

Binary relation indicated with $\rightarrow^*$ which is the reflexive and transitive closure of the small-step evaluation.

Note that the transitive rule is a single step and a multi step, although it can be proved that it allows concatenation of single steps.

1.1 Properties of programs

- Determinism: suppose that for any given configuration, if this can step in different ways these are equal
- Termination: evaluation terminates. Formally, for the multi step there is a configuration that it can multi step to without other possible steps.
- Soundness: if we run a program long enough we should get a final integer, but it's not true because stores are partial function and if a variable isn't defined we're cooked.

Definition 1.1.1: Free Variables

Given the function $fvs(e)$ defined inductively as:

- If e is a variable x , then $fvs(e) = \{x\}$.
- If e is a constant, then $fvs(e) = \emptyset$.
- If e is $e_1 + e_2$, then $fvs(e) = fvs(e_1) \cup fvs(e_2)$.
- If e is $x := e_1$, then $fvs(e) = fvs(e_1)$.
- If e is a sequence $e_1; e_2$, then $fvs(e) = fvs(e_1) \cup fvs(e_2)$.

Note that in $x := x + 1; y := x$ we still need to add x to the free variables because it's in the rhs of the first assignment.

Definition 1.1.2: Well-Formedness

For any environment σ and expression e , if all free variables in e are in σ , then e is well-formed under σ .

Two properties that imply soundness:

Definition 1.1.3: Progress

Every well-formed configuration is either an `Int` or can step to another configuration.

Definition 1.1.4: Preservation

If a configuration is well-formed, any following configuration it steps to will also be well-formed.

with these lemmas and termination we can show that a well-formaed configuration is sound.

1.1.1 Inductive Sets

A bit of maths before the proof

Definition 1.1.5: Inductive Set

A set S is **inductive** if:

1. $\emptyset \in S$, and
2. for every set x , if $x \in S$, then $x \cup \{x\} \in S$.

We have used inductive setts with the inductive rules (remember that $\rightarrow$ is a relation, so a set of tuples) and BNF grammar, free-variables and multistep. Another example is the natural numbers using the successor definition.

1.1.2 Inductive Proofs

If we want to prove something about an inductive set (that can be infinite by closing under the inductive rules) we can use the induction principle.

The base cases are all the axioms.

Sometimes the properties are defined over multiple inductive setts, so we need to chose one.